

Vivian Y Zhu | Data Scientist

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SUMMARY

Second-year Data Science student with strong experience in Python, SQL, and Git, and a solid foundation in data cleaning, analysis, and workflow management. Detail-oriented and collaborative, with hands-on experience translating complex datasets into actionable insights and building reproducible data pipelines. Eager to contribute to data-driven decision-making, apply analytical techniques, and learn industry best practices in data science.

EDUCATION

University of California, San Diego

Bachelor of Science in Data Science | GPA: 3.9

La Jolla, California

Expected June 2028

SKILLS

Programming: Python, SQL, Java, HTML, CSS, R, C, MATLAB, scikit-learn, pandas, numpy, matplotlib, seaborn

Software Tools: Git/Github, Excel, Tableau, Jupyter Notebook, IntelliJ, Streamlit

EXPERIENCE

Queen's University Belfast

Research Intern

Belfast, United Kingdom

July 2025 - August 2025

- Enabled resilience analysis in 100+ countries by integrating 1M+ supply chain rows and 4K+ disaster records over 10+ years into a visualization platform (Flask, Pandas, Tableau).
- Automated country-level dashboards that generated 10+ interactive visualizations on demand, reducing manual analysis time by 60% and guiding disaster-response research.
- Joined, cleaned, and filtered disparate datasets to uncover relationships between natural disasters and supply chain vulnerability, supporting future research direction.

PROJECTS

Stackbot | *Python, scikit-learn, NLP, Transformers (Hugging Face), Cosine Similarity, Streamlit*

- Collaborated on building a closed-corpus chatbot using 600K+ Python questions and 1M+ answers, avoiding unrestricted external APIs to improve data security and grounded strictly in Stack Overflow data to reduce hallucinations
- Designed a retrieval-first QA system using transformer embeddings and cosine similarity, ranking answers with TF-IDF similarity scores and answer-question upvote ratios.
- Optimized data storage and cloud deployment by tuning embedding representations and resource constraints, reducing memory usage by ~20% while maintaining retrieval accuracy.

Java Custom DataFrame | *Java, Data Structures, Algorithm Optimization, OOP, Data Manipulation, IntelliJ*

- Engineered a column-oriented DataFrame in Java to handle mixed-type datasets with 680+ rows, enabling efficient manipulation and exploration of large datasets for analytics pipelines.
- Optimized data processing by replacing linked lists with binary trees, reducing operation time from $O(n)$ to $O(\log n)$
- Implemented 10+ data manipulation methods mirroring Pandas functionality, facilitating reusable and scalable data pipelines.

Song Recommendation & Analysis System | *Python, pandas, numpy, matplotlib, seaborn, Data Visualization*

- Developed a music recommender system analyzing 200+ songs across 7 audio features from a 1B+ stream dataset to identify similar tracks through feature-based similarity.
- Built a lyric search tool returning context-based results (target line + 2 surrounding lines) and created 10+ interactive visualizations exploring song features and trends for deeper musical insights.

LEADERSHIP

Data Science Student Society @ UC San Diego

Procurement Chair

La Jolla, California

October 2024 - Present

- Streamlined procurement by managing 100+ vendor relationships, coordinating cross-team merchandise delivery for 200+ attendee events, and strengthening community engagement through efficient operations.

Campus Outreach Lead

October 2024- Present

- Initiated and coordinated collaborations with campus ambassadors and student organizations to support joint projects and events.
- Fostered partnerships with startups and professionals, communicating across teams to secure sponsorships and speakers, strengthen event coordination, and expand DS3's industry presence.